

DAY 3: ORDINARY DIFFERENTIAL EQUATIONS

- NOTES

THE LANGUAGE OF DYNAMICS

Calculus Bootcamp for Physics Students

Goal

Understand Ordinary Differential Equations (ODEs) including separable equations, first-order linear equations, and second-order linear equations with constant coefficients. Learn how these equations describe physical phenomena in mechanics, circuits, and oscillations.

1 WHAT IS A DIFFERENTIAL EQUATION?

A differential equation is an equation that relates a function to its derivatives. It describes how a quantity changes in time or space.

1.1 Examples in Physics

- **Newton's Second Law:** $m \frac{d^2x}{dt^2} = F(x, v, t)$ - describes motion under forces
- **Radioactive Decay:** $\frac{dN}{dt} = -\lambda N$ - describes how atoms decay
- **RL Circuit:** $L \frac{dI}{dt} + RI = V(t)$ - describes current in a circuit
- **Simple Harmonic Oscillator:** $\frac{d^2x}{dt^2} + \omega^2 x = 0$ - describes springs and pendulums

1.2 Order of a Differential Equation

The order of a differential equation is the highest derivative that appears.

- **First-order:** $\frac{dy}{dx} = ky$ (radioactive decay)

- **Second-order:** $\frac{d^2x}{dt^2} + \omega^2x = 0$ (harmonic oscillator)

1.3 Ordinary vs. Partial

- **Ordinary Differential Equation (ODE):** Involves derivatives with respect to a single independent variable.
- **Partial Differential Equation (PDE):** Involves partial derivatives with respect to multiple variables.

In these notes, we focus on ODEs.

2 FIRST-ORDER DIFFERENTIAL EQUATIONS

2.1 Separation of Variables

A first-order ODE is separable if it can be written in the form:

$$\boxed{\frac{dy}{dx} = f(x)g(y)}$$

2.1.1 Solution Method

1. Separate the variables so that all y terms are on one side and all x terms on the other:

$$\frac{dy}{g(y)} = f(x) dx$$

2. Integrate both sides:

$$\int \frac{dy}{g(y)} = \int f(x) dx + C$$

3. Solve for y explicitly if possible. If not, the implicit solution is acceptable.

2.1.2 Physical Examples

Radioactive Decay:

$$\frac{dN}{dt} = -\lambda N$$

where $N(t)$ is the number of radioactive atoms and λ is the decay constant.

Separating variables:

$$\frac{dN}{N} = -\lambda dt$$

Integrating:

$$\ln |N| = -\lambda t + C$$

Solving for N :

$$N(t) = N_0 e^{-\lambda t}$$

where $N_0 = N(0)$ is the initial number of atoms.

Newton's Law of Cooling:

$$\frac{dT}{dt} = -k(T - T_{\text{env}})$$

where $T(t)$ is the temperature of an object, T_{env} is the constant environmental temperature, and $k > 0$ is a constant.

Let $u = T - T_{\text{env}}$. Then:

$$\frac{du}{dt} = -ku$$

This is the same form as radioactive decay, so:

$$u(t) = u_0 e^{-kt}$$

where $u_0 = T(0) - T_{\text{env}}$. Therefore:

$$T(t) = T_{\text{env}} + (T(0) - T_{\text{env}})e^{-kt}$$

Terminal Velocity with Air Resistance:

$$m \frac{dv}{dt} = mg - kv$$

where m is mass, g is gravity, k is the drag coefficient, and v is velocity.

Separating variables:

$$\frac{dv}{g - \frac{k}{m}v} = dt$$

Integrating (using substitution $u = g - \frac{k}{m}v$):

$$-\frac{m}{k} \ln \left| g - \frac{k}{m}v \right| = t + C$$

Solving for v :

$$v(t) = \frac{mg}{k} + \left(v_0 - \frac{mg}{k} \right) e^{-(k/m)t}$$

As $t \rightarrow \infty$, $v \rightarrow \frac{mg}{k}$, the terminal velocity.

2.2 First-Order Linear ODEs

A first-order linear ODE has the standard form:

$$\boxed{\frac{dy}{dx} + P(x)y = Q(x)}$$

2.2.1 Solution Method: Integrating Factor

1. Compute the integrating factor:

$$\mu(x) = e^{\int P(x) dx}$$

(Note: The constant of integration can be omitted when finding μ .)

2. Multiply both sides of the ODE by $\mu(x)$:

$$\mu(x) \frac{dy}{dx} + \mu(x)P(x)y = \mu(x)Q(x)$$

3. Recognize that the left side is the derivative of $\mu(x)y$:

$$\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$$

This works because $\frac{d\mu}{dx} = \mu P$.

4. Integrate both sides with respect to x :

$$\mu(x)y = \int \mu(x)Q(x) dx + C$$

5. Solve for y :

$$y = \frac{1}{\mu(x)} \left[\int \mu(x)Q(x) dx + C \right]$$

2.2.2 Physical Examples

RL Circuit:

$$L \frac{dI}{dt} + RI = V$$

where $I(t)$ is current, L is inductance, R is resistance, and V is constant voltage.

Rewrite in standard linear form:

$$\frac{dI}{dt} + \frac{R}{L}I = \frac{V}{L}$$

Here $P(t) = \frac{R}{L}$ and $Q(t) = \frac{V}{L}$.

Integrating factor:

$$\mu(t) = e^{\int \frac{R}{L} dt} = e^{(R/L)t}$$

Multiply the ODE by μ :

$$\frac{d}{dt} [Ie^{(R/L)t}] = \frac{V}{L} e^{(R/L)t}$$

Integrate:

$$Ie^{(R/L)t} = \frac{V}{L} \cdot \frac{L}{R} e^{(R/L)t} + C = \frac{V}{R} e^{(R/L)t} + C$$

Solve for I :

$$I(t) = \frac{V}{R} + Ce^{-(R/L)t}$$

If the initial current is $I(0) = I_0$, then $C = I_0 - \frac{V}{R}$, so:

$$I(t) = \frac{V}{R} + \left(I_0 - \frac{V}{R}\right) e^{-(R/L)t}$$

The quantity $\tau = L/R$ is the **time constant** - the time for the current to change by a factor of e .

RC Circuit:

$$R \frac{dQ}{dt} + \frac{Q}{C} = V$$

where $Q(t)$ is charge on the capacitor.

Rewrite:

$$\frac{dQ}{dt} + \frac{1}{RC} Q = \frac{V}{R}$$

This has the same form as the RL circuit. For charging ($Q(0) = 0$):

$$Q(t) = CV(1 - e^{-t/RC})$$

For discharging ($V = 0$, $Q(0) = Q_0$):

$$Q(t) = Q_0 e^{-t/RC}$$

3 SECOND-ORDER LINEAR ODES WITH CONSTANT COEFFICIENTS

3.1 Superposition for Homogeneous Linear Equations

For a **homogeneous** linear equation ($f(t) = 0$), if $y_1(t)$ and $y_2(t)$ are solutions, then:

$$y(t) = C_1 y_1(t) + C_2 y_2(t) \text{ is also a solution}$$

for any constants C_1 and C_2 .

3.2 General Form

$$a \frac{d^2 y}{dt^2} + b \frac{dy}{dt} + cy = f(t)$$

If $f(t) = 0$, the equation is **homogeneous**. Otherwise, it's **non-homogeneous**. These equations appear throughout physics:

- Mass on a spring: $m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = F(t)$
- LCR circuit: $L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = V(t)$

3.3 Homogeneous Equations ($f(t) = 0$)

$$a \frac{d^2 y}{dt^2} + b \frac{dy}{dt} + cy = 0$$

3.3.1 Method: Characteristic Equation

Assume a solution of the form $y = e^{rt}$. Substitute into the ODE:

$$ar^2 e^{rt} + br e^{rt} + ce^{rt} = 0$$

Since $e^{rt} \neq 0$, we can divide by it:

$$ar^2 + br + c = 0$$

This is the **characteristic equation**. It is a quadratic equation whose roots determine the form of the solution.

3.4 Three Cases for the Roots

3.4.1 Case 1: Real and Distinct Roots ($r_1 \neq r_2$)

The general solution is:

$$y(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

Example: $y'' - 3y' + 2y = 0$

Characteristic: $r^2 - 3r + 2 = 0 \Rightarrow (r - 1)(r - 2) = 0 \Rightarrow r = 1, 2$

Solution: $y(t) = C_1 e^t + C_2 e^{2t}$

3.4.2 Case 2: Repeated Real Roots ($r_1 = r_2 = r$)

When the discriminant is zero, we get a double root. The general solution is:

$$y(t) = (C_1 + C_2 t) e^{rt}$$

The factor t provides a second linearly independent solution.

Example: $y'' + 6y' + 9y = 0$

Characteristic: $r^2 + 6r + 9 = 0 \Rightarrow (r + 3)^2 = 0 \Rightarrow r = -3$ (repeated)

Solution: $y(t) = (C_1 + C_2 t) e^{-3t}$

3.4.3 Case 3: Complex Conjugate Roots ($r = \alpha \pm i\beta$)

When the discriminant is negative, the roots are complex:

$$r = \alpha \pm i\beta, \quad \text{where } \alpha = -\frac{b}{2a}, \quad \beta = \frac{\sqrt{4ac - b^2}}{2a}$$

The general solution is:

$$y(t) = e^{\alpha t} (C_1 \cos \beta t + C_2 \sin \beta t)$$

This represents oscillations that grow ($\alpha > 0$), decay ($\alpha < 0$), or persist ($\alpha = 0$).

Example: $y'' + 4y = 0$

Characteristic: $r^2 + 4 = 0 \Rightarrow r = \pm 2i$

Here $\alpha = 0$, $\beta = 2$, so:

$$y(t) = C_1 \cos 2t + C_2 \sin 2t$$

3.5 Physical Examples of Homogeneous Second-Order ODEs

3.5.1 Simple Harmonic Oscillator

A mass m on a spring with spring constant k (no damping) satisfies:

$$m \frac{d^2 x}{dt^2} + kx = 0$$

Divide by m :

$$\frac{d^2 x}{dt^2} + \omega_0^2 x = 0, \quad \text{where } \omega_0 = \sqrt{\frac{k}{m}}$$

The characteristic equation is $r^2 + \omega_0^2 = 0$, so $r = \pm i\omega_0$.

The general solution is:

$$x(t) = A \cos \omega_0 t + B \sin \omega_0 t$$

This can also be written as:

$$x(t) = C \cos(\omega_0 t - \phi)$$

where $C = \sqrt{A^2 + B^2}$ is the amplitude and $\phi = \arctan(B/A)$ is the phase constant.

The period of oscillation is $T = \frac{2\pi}{\omega_0}$.

3.5.2 Damped Harmonic Oscillator

With damping proportional to velocity:

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = 0$$

Divide by m :

$$\frac{d^2 x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_0^2 x = 0$$

where $\beta = \frac{b}{2m}$ (damping coefficient) and $\omega_0 = \sqrt{\frac{k}{m}}$ (natural frequency).

The characteristic equation is:

$$r^2 + 2\beta r + \omega_0^2 = 0$$

The roots are:

$$r = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$$

Three cases arise:

- **Underdamped** ($\beta < \omega_0$): Complex conjugate roots

$$r = -\beta \pm i\omega_1, \quad \omega_1 = \sqrt{\omega_0^2 - \beta^2}$$

The solution is:

$$x(t) = e^{-\beta t}(A \cos \omega_1 t + B \sin \omega_1 t)$$

This represents oscillations with exponentially decaying amplitude.

- **Critically Damped** ($\beta = \omega_0$): Repeated real roots $r = -\beta$

$$x(t) = (A + Bt)e^{-\beta t}$$

This returns to equilibrium fastest without oscillating.

- **Overdamped** ($\beta > \omega_0$): Real distinct roots $r_1, r_2 < 0$

$$x(t) = Ae^{r_1 t} + Be^{r_2 t}$$

This returns to equilibrium slowly without oscillating.

3.6 Non-Homogeneous Equations

$$a \frac{d^2 y}{dt^2} + b \frac{dy}{dt} + cy = f(t)$$

3.6.1 General Solution Structure

The general solution is the sum of the homogeneous solution and a particular solution:

$$y(t) = y_h(t) + y_p(t)$$

where:

- $y_h(t)$ solves the homogeneous equation ($f(t) = 0$)
- $y_p(t)$ is any function that satisfies the original non-homogeneous equation

3.7 Method of Undetermined Coefficients

This method works when $f(t)$ has a specific form: polynomials, exponentials, sines, cosines, or products of these.

3.7.1 Procedure

1. Find the homogeneous solution $y_h(t)$.
2. Make an initial guess for $y_p(t)$ based on the form of $f(t)$.
3. Substitute the guess into the original ODE and solve for the unknown coefficients.
4. Add $y_h + y_p$ to get the general solution.

3.7.2 Table of Guesses for y_p

$f(t)$	Initial Guess for y_p
$P_n(t)$ (polynomial degree n)	$A_n t^n + A_{n-1} t^{n-1} + \cdots + A_0$
$e^{\alpha t}$	$A e^{\alpha t}$
$\sin \beta t$ or $\cos \beta t$	$A \sin \beta t + B \cos \beta t$
$e^{\alpha t} \sin \beta t$	$e^{\alpha t} (A \sin \beta t + B \cos \beta t)$
$t^n e^{\alpha t}$	$e^{\alpha t} (A_n t^n + \cdots + A_0)$

4 SUMMARY OF SOLUTION METHODS

Type	Standard Form	Solution Method
Separable First-Order	$\frac{dy}{dx} = f(x)g(y)$	Separate and integrate
Linear First-Order	$\frac{dy}{dx} + P(x)y = Q(x)$	Integrating factor $\mu = e^{\int P dx}$
Homogeneous Second-Order	$ay'' + by' + cy = 0$	Characteristic equation $ar^2 + br + c = 0$
Non-Homogeneous Second-Order	$ay'' + by' + cy = f(t)$	$y = y_h + y_p$ (undetermined coefficients)

5 CHARACTERISTIC EQUATION CASES SUMMARY

Roots	General Solution	Physical Interpretation
Real distinct r_1, r_2	$y = C_1 e^{r_1 t} + C_2 e^{r_2 t}$	Exponential growth ($r > 0$) or decay ($r < 0$)
Repeated real r	$y = (C_1 + C_2 t) e^{rt}$	Critical damping
Complex $r = \alpha \pm i\beta$	$y = e^{\alpha t} (C_1 \cos \beta t + C_2 \sin \beta t)$	Oscillations with growth/decay

6 KEY PHYSICAL MODELS

6.1 Radioactive Decay (First-Order)

$$\frac{dN}{dt} = -\lambda N \quad \Rightarrow \quad N(t) = N_0 e^{-\lambda t}$$

6.2 Newton's Law of Cooling (First-Order)

$$\frac{dT}{dt} = -k(T - T_{\text{env}}) \quad \Rightarrow \quad T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$$

6.3 RL Circuit (First-Order Linear)

$$L \frac{dI}{dt} + RI = V \quad \Rightarrow \quad I(t) = \frac{V}{R} + \left(I_0 - \frac{V}{R} \right) e^{-(R/L)t}$$

6.4 RC Circuit (First-Order Linear)

$$\text{Charging: } Q(t) = CV(1 - e^{-t/RC}), \quad \text{Discharging: } Q(t) = Q_0 e^{-t/RC}$$

6.5 Simple Harmonic Oscillator (Second-Order Homogeneous)

$$\frac{d^2x}{dt^2} + \omega_0^2 x = 0 \quad \Rightarrow \quad x(t) = A \cos \omega_0 t + B \sin \omega_0 t$$

6.6 Damped Harmonic Oscillator (Second-Order Homogeneous)

$$\frac{d^2x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_0^2 x = 0$$

- Underdamped ($\beta < \omega_0$): $x(t) = e^{-\beta t}(A \cos \omega_1 t + B \sin \omega_1 t)$
- Critically damped ($\beta = \omega_0$): $x(t) = (A + Bt)e^{-\beta t}$
- Overdamped ($\beta > \omega_0$): $x(t) = Ae^{r_1 t} + Be^{r_2 t}$

6.7 Driven Harmonic Oscillator (Second-Order Non-Homogeneous)

$$\frac{d^2x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_0^2 x = F_0 \cos \omega t$$

At resonance ($\omega = \omega_0$), amplitude grows linearly with time.

Important Constants and Parameters

$$\tau = \frac{L}{R} \quad (\text{RL circuit time constant})$$

$$\tau = RC \quad (\text{RC circuit time constant})$$

$$\omega_0 = \sqrt{\frac{k}{m}} \quad (\text{natural frequency of spring-mass system})$$

$$\beta = \frac{b}{2m} \quad (\text{damping coefficient})$$

$$\omega_1 = \sqrt{\omega_0^2 - \beta^2} \quad (\text{damped frequency})$$